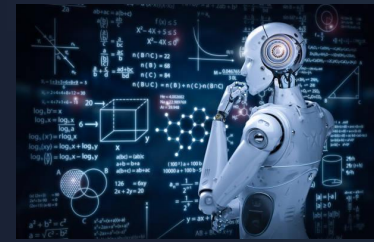




RESEARCH PRESENTATION



Comparative Analysis of Dimensionality Reduction Methods

Evaluating Impact on Image Classification Performance

Sahil Dadhwal (917598320)

Purva Khadke (924112595)

Sarvesh Halbe (925581073)

Introduction & Objective

The Problem

- High dimensional image processing creates massive compositional bottlenecks
- Goal: To evaluate dimensionality reduction to balance speed vs. accuracy

The Objective

- Compare Linear (PCA) vs. Nonlinear (UMAP, Autoencoder) efficacy in preserving class-discriminative features.
- Determine if Transformers adapt better to compressed data than traditional Autoencoders.
- Identify the "sweet spot" ratio where computational efficiency is gained without critical accuracy loss.

Methodology

Experiment Design

CIFAR-100 Overview

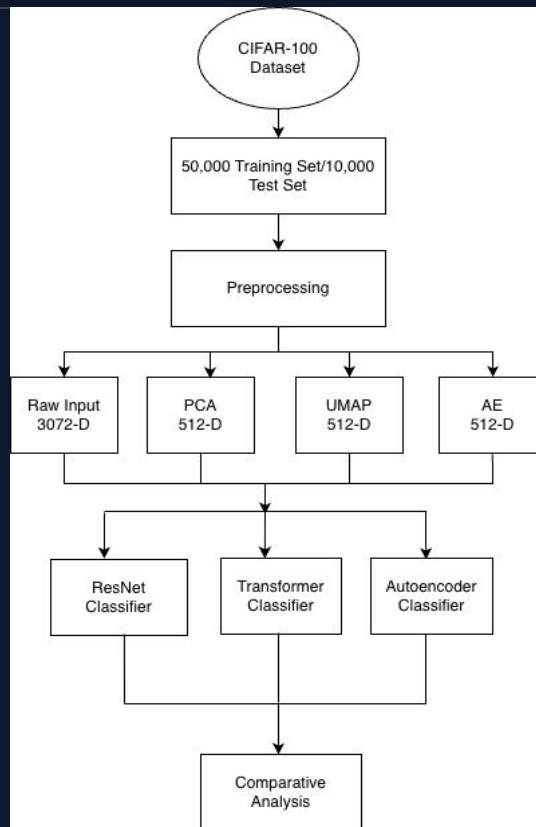
- **Volume:** 60,000 total images (50,000 Train / 10,000 Test).
- **Complexity:** 100 distinct classes; 32 x 32 pixel resolution.

Standard Preprocessing

- **Flattening:** Converted 32 x 32 x 3 images → 3,072-dimensional vectors.
- **Normalization:** Scaled pixel values from [0, 255] to [0, 1] for convergence.

Dimensionality Reduction Protocol

- **Target:** Reduced features from 3,072-D → 512-D.
- **Leakage Prevention:** Reducers (PCA, UMAP, AE) fitted **exclusively** on Training Set.
- Learned transformations applied subsequently to Test Set



Methodology

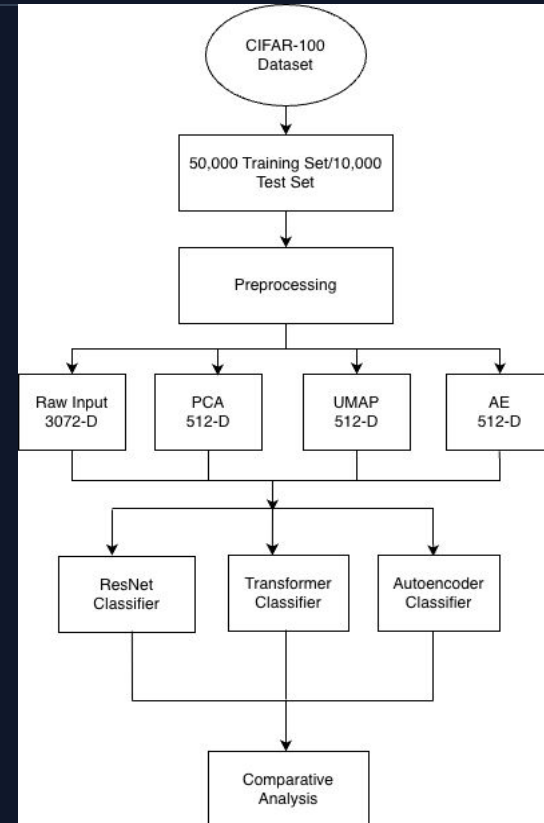
Dataset & Preprocessing

Experimental Conditions

- **4 Input Types:**
 - **Raw:** Baseline (3,072-D).
 - **PCA:** Linear reduction (95.23% variance retained).
 - **UMAP:** Nonlinear manifold projection.
 - **Autoencoder:** Learned nonlinear encoding.
- **3 Classifiers:** ResNet (Frozen), Transformer, Autoencoder Classifier.

Key Hyperparameters

- **Target Dimensions:** Fixed at **512-D** for all reduction methods.
- **Training:** 30 Epochs (reduced inputs) vs. 10 Epochs (raw fine-tuning).
- **Optimization:** Adam/AdamW optimizer; Batch size of 128.
- **AE Specifics:** Loss weighted 70% for classification, 30% for reconstruction



Reduction Techniques Evaluated



PCA

Principal Component Analysis

Linear technique. Projects data onto axes of greatest variance. Reduced from 3,072 to 512 dims (95% variance).



UMAP

Uniform Manifold Approx.

Nonlinear manifold learning. Optimizes to preserve local neighborhood structure. Typically used for visualization.

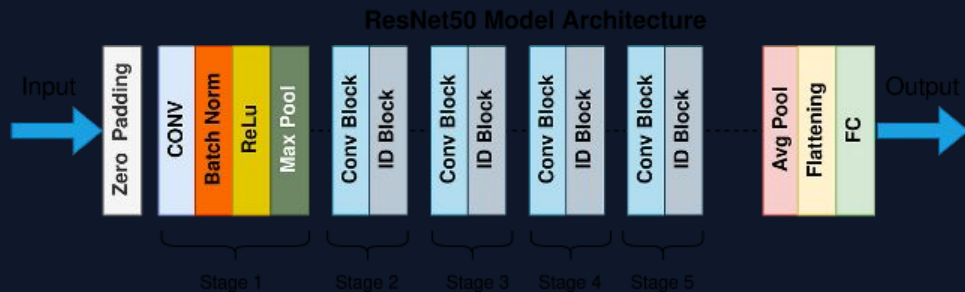


Autoencoder

Neural Compression

Symmetric neural network. Learns a compressed latent space representation via reconstruction loss.

Classifier Architectures



1. ResNet-50

Deep residual network using transfer learning (ImageNet). For reduced features, an MLP is used as spatial structure is lost.

2. Transformer

Treats image patches as tokens with self-attention. Trained from scratch (4 layers, 8 heads).

3. Autoencoder Classifier

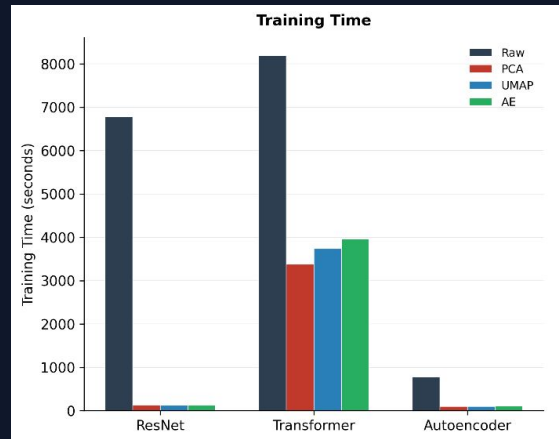
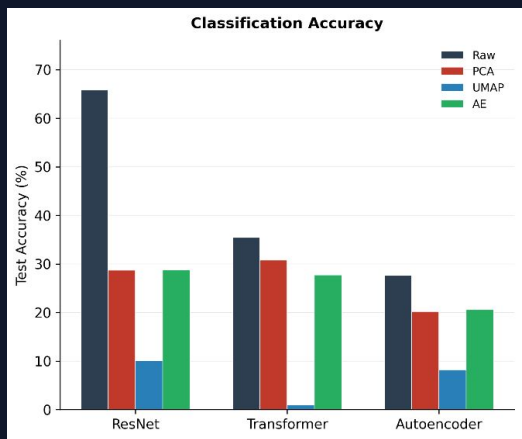
Hybrid model. Learns features via reconstruction while simultaneously classifying via a bottleneck head.

Classification Accuracy Results

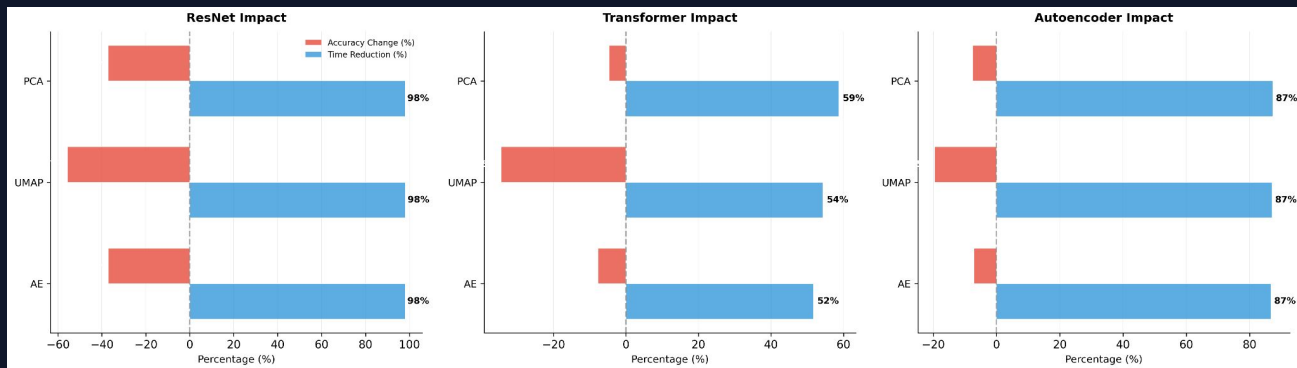
Model	Raw	PCA	UMAP	AE
ResNet	65.82%	28.69%	10.07%	28.75%
Transformer	35.45%	30.79%	1.00%	27.74%
Autoencoder	27.68%	20.22%	8.20%	20.57%
Average	42.98%	26.57%	6.42%	25.69%

(Raw features with ResNet achieved the highest accuracy. UMAP performed near random chance.)

Performance Comparison



(Transformer only reached 35.45%, showing the importance of pretraining. UMAP performed near random chance)



Efficiency vs. Accuracy Trade-off

98%

Faster Training

Reduction in training time for ResNet when using reduced features compared to raw data.

56%

Accuracy Loss

The cost of efficiency: A massive drop in predictive performance for the ResNet model.

~26%

Linear Baseline

PCA (Linear) performed comparably to Autoencoders (Nonlinear), suggesting 32x32 images rely on simple features.

PCA vs Autoencoder

26.57%

PCA

Similar?

Autoencoder trained without class labels, so it learned to reconstruct, not classify. Simple linear PCA worked just as well."

25.69%

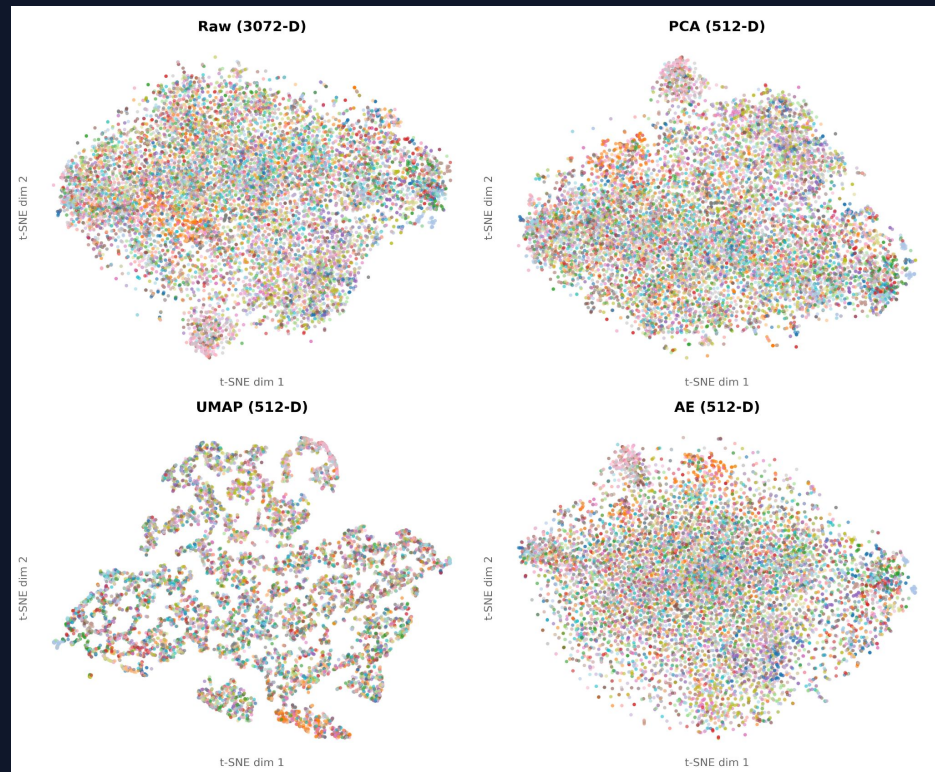
AutoEncoder

Feature Space Analysis

Why did UMAP fail?

Visual inspection via t-SNE reveals critical insights:

- **Fragmented Topology:** UMAP produced disjointed, ring-like clusters rather than cohesive class groupings.
- **Neighborhood vs. Global:** UMAP optimizes for local neighbors, which may not align with class separability in high dimensions.
- **PCA/AE Structure:** While less accurate than raw data, PCA and Autoencoders maintained a more continuous feature space.



Conclusion & Key Takeaways

1. Original Data is Best

Unaltered images yielded significantly better results (**65.8%**) than any simplified version. Modern CNNs (ResNet) excel at extracting features directly from raw pixels.

2. Linear vs. Nonlinear

Simple linear reduction (**PCA - 28.69%**) performed as well as complex nonlinear methods (**Autoencoders - 28.75%**) for this task, suggesting added complexity does not always yield better features.

3. UMAP Limitations

While UMAP is great for making visual charts, it failed as a tool for model training, as the accuracy dropped below **10%**.

Therefore, we conclude that PCA is the practical choice for strictly resource-constrained environments where operational speed is prioritized over predictive precision.

Thank
You